Predicting Land Prices in the Colombo District

IT5514 Applied Machine Learning - Assignment 02

Module: IT5514 Applied Machine Learning | **Assignment 02** | **Individual Assignment**

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1. Introduction

Land is the largest purchase most Sri Lankan families ever make, yet there is no public, plot-level price index for the Colombo district. Buyers and sellers mostly judge prices from a handful of advertisements in the same area, and asking prices for similar land can differ several times over. The aim of this assignment was to build a model that estimates a fair **asking price per perch** for a plot of land in the Colombo district, from its area, size, land type and the facilities around it.

The work follows the assignment guideline:

1. **Problem and dataset:** a real local problem and a real Sri Lankan dataset, with ethical data use (Sections 1-3).
2. **A new algorithm: NGBoost (Natural Gradient Boosting)**, which was not covered in the lectures, was selected and justified (Section 4.3). Unlike the taught models it predicts a whole probability distribution, which gives a **price range** for every plot. It was compared with the taught models (Ridge regression, Random Forest, XGBoost, LightGBM, CatBoost and a stacked ensemble).
3. **Training and evaluation:** train / validation / test split, tuned hyperparameters, and appropriate metrics (Sections 4-5).
4. **Explainability:** permutation feature importance, SHAP, LIME and partial dependence plots, including whether the model's behaviour matches domain knowledge (Section 6).
5. **Critical discussion:** limitations, data quality, risks of bias and ethical considerations (Section 8).
6. **Front-end (bonus):** a deployed web application where users enter a plot and see its predicted price **and an explanation** of it. The project is also published to GitHub with an automated MLOps pipeline (Section 7).

The main task is **regression** (predicting the price). As an extension, the same problem was also framed as **classification** into Low, Medium and High price bands.

All the work was done in Python with Pandas, scikit-learn, XGBoost, LightGBM, CatBoost and NGBoost. The accompanying files are:

File	Contents
`01_data_preparation.ipynb`	Cleaning and preprocessing (Section 3)
`02_modelling.ipynb`	Models, tuning, comparison, interpretation, classification, inference (Sections 4-7)
`03_unseen_areas.ipynb`	Check of how well the models work for areas they have never seen (Section 6.3)
`04_explainability.ipynb`	SHAP, LIME, CatBoost SHAP and partial dependence plots (Sections 6.4-6.5)
`05_ngboost.ipynb`	NGBoost, the new algorithm: tuning, accuracy, price-range calibration, SHAP (Sections 4.3, 6.7)
`app.py`, `explain.py`, `catboost_wrapper.py`, `ngboost_wrapper.py`	Front-end application with explanations and price ranges (Section 7)
`land_price_model.pkl`	Saved final models

2. Dataset

Source. *Land prices of Colombo district*, published on Kaggle by user ruchiraayeshmantha (https://www.kaggle.com/datasets/ruchiraayeshmantha/land-prices-of-colombo-district). The dataset contains land sale advertisements from the Colombo district, posted between **31 July and 21 September 2021**. Each advertisement was enriched with location information: the distance to Colombo Fort, and the distance to and number of nearby schools, universities, hospitals, banks, finance companies, supermarkets, fuel stations, railway stations and expressway entrances.

The file has **4,494 rows and 35 columns** with **no missing values**:

Group	Columns
Listing	`Address` (area name, 160 areas), `Land_size(Perches)`, `Land_type`, `Posted_Date_new`
Price	`Mentioned Price(Rs)`, `Price_Scale`, `Price per Perch`
Location	27 distance and count features (e.g. `Distance from fort`, `min_dist_nearest_railway`, `count_banks_within_2km`)
Other	`Address_ID` (row identifier)

The busiest areas are Ranala (350 listings), Makandana (270), Kaduwela (250), Kesbewa (206) and Piliyandala (204). These are suburban areas where large blocks of land are divided and sold plot by plot.

Ethical data use. The data comes from publicly posted advertisements and was obtained from a public Kaggle dataset. It contains **no personal information**: no seller names, phone numbers, email addresses or exact property addresses. The location is recorded only at the level of the area (e.g. "Battaramulla"), and the location features were derived from public places such as schools, banks and hospitals. The data is used only for this academic project, and the source is credited.

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3. Data Preparation

3.1 Choice of target

Two columns describe the price: `Mentioned Price(Rs)` (the number written in the advertisement) and `Price per Perch`. **Price per perch** was chosen as the target, because it makes plots of different sizes comparable. A 6-perch plot in Nugegoda and a 40-perch plot in Kaduwela can't be compared on total price, but they can be compared per perch, and this is also how land is normally advertised and discussed in Sri Lanka.

`Mentioned Price(Rs)` was then **removed from the features**. For most rows it is either identical to the target or equal to the target multiplied by the land size, so leaving it in would hand the model the answer (data leakage). `Address_ID` was also removed, since it is just a row number.

3.2 Price scale

Advertisers quote prices in different ways, recorded in `Price_Scale`: per perch (3,568 rows), as a total price (923 rows), or per acre (3 rows). Checking `Price per Perch` against the mentioned price for each scale showed that:

- **per perch:** `Price per Perch` equals the mentioned price, which is correct
- **total price:** `Price per Perch` equals the mentioned price ÷ land size, so these were already converted correctly
- **per acre:** `Price per Perch` still held the **per-acre** price, which is wrong

The three per-acre rows were corrected by dividing by 160 (1 acre = 160 perches). After that, `Price_Scale` only describes how the advertisement was written, not the land, so it was dropped.

3.3 Duplicates

After removing the ID and the mentioned price, **198 rows were exact duplicates** and were removed (4,296 rows remained). This is more than the 177 duplicates in the raw file. The

difference is listings that were posted twice, once as a per-perch price and once as a total price: they become identical once the price is expressed per perch. Duplicates were removed **before** splitting the data, so the same listing could not end up in both the training and the test set.

3.4 Skewed target and outliers

The price per perch ranges from about Rs 4,600 to Rs 85 million, with a median of about Rs 900,000. The distribution is extremely right-skewed (skewness 7.19), as Figure 1 shows.

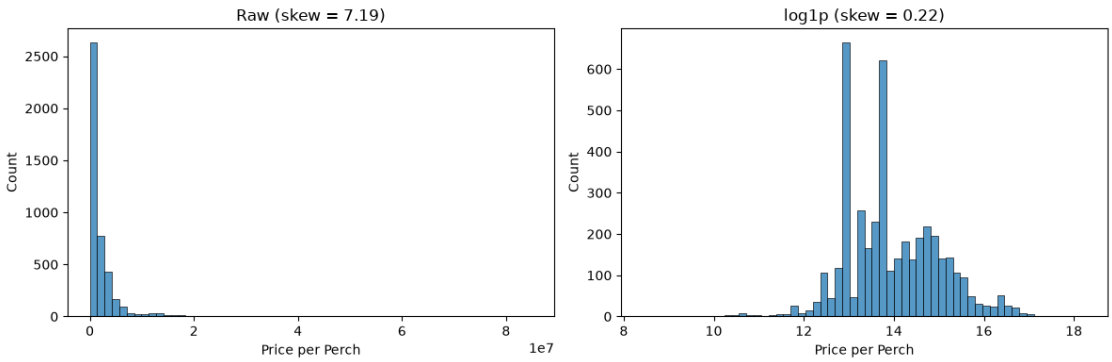


Figure 1: Price per perch before and after the log transform

Applying the IQR rule from the module's feature-engineering lab directly to this target would flag 297 listings as outliers. Most of them are genuine, expensive land close to the city centre, so removing them would remove exactly the information the model needs about central Colombo. Instead:

1. The target was **log-transformed** with `log1p`, which reduces the skewness to 0.22.
2. The IQR rule was applied on the **log scale**, which flags only **21 listings**. These are extreme values such as a few thousand rupees per perch (almost certainly typos or a misread price scale), and they were removed. **4,275 rows** remained.
3. All models predict the log price. Predictions are converted back to rupees with `expm1`.

Outliers in the features (distances and counts) were left alone, because tree-based models split on thresholds and are not distorted by extreme feature values the way linear models are.

The two tall peaks in the log histogram come from popular round asking prices (for example Rs 400,000, 430,000 and 450,000 per perch). They are mostly in suburban areas such as Kaduwela, Homagama and Bomiriya, where developers sell many plots from the same block at the same price.

3.5 Land type

`Land_type` combines several types in one value, for example "Commercial, Residential, Other", giving 14 different combinations, some with only 1-3 listings. Since a plot can

legitimately be several types at once, it was split into four yes/no columns: `is_Residential`, `is_Commercial`, `is_Agricultural` and `is_Other`. This way a "Commercial, Residential" plot shares information with all other commercial plots and all other residential plots, instead of being a rare category of its own.

3.6 Posting date and area

- **Posting date** was dropped. All listings were posted within about eight weeks, which is too short for land prices to change meaningfully, so the date would mostly add noise.
- **Area (`Address`)** has 160 values. One-hot encoding would add 160 sparse columns, so it was **target encoded** instead: each area is replaced by its average log price, using scikit-learn's `TargetEncoder`. The encoder uses internal cross-fitting, so a listing's own price is never used to encode it. CatBoost handles the area natively (Section 4.4).
- **Scaling** was applied only for the linear baseline (Ridge regression), whose penalty treats all coefficients equally and therefore needs features on comparable scales. Tree-based models don't need scaling, because a split on "distance < 5 km" works the same whatever the units.

3.7 Location features are recorded per area, not per plot

While interpreting the results, an important property of the data came to light: **all 27 location features have exactly one value per area.** The distances and counts were calculated from the area name, not from each plot's position, so every listing in, say, Kaduwela has the same distance to Fort, the same number of nearby banks, and so on. The whole dataset contains only 160 distinct location "points", one for each area.

So, within an area, the only things that differ between listings are the land size, the land type and the price itself. This has three consequences, discussed in Sections 6.1, 6.3 and 7: the feature importances must be read carefully, a separate test on unseen areas is needed, and the application should not let users edit the location values.

3.8 Exploratory analysis of the cleaned data

Two questions were explored before modelling: how much price depends on the area, and which features move with price and with each other.

Price by area. The 160 areas are very unevenly represented. The median area has only 8 listings, and 88 areas have fewer than 10. Among the 58 areas with at least 15 listings, the median price per perch ranges from about **Rs 215,000 in Kosgama to Rs 18 million in Kurunduwatta, an 84-fold difference** (Figure 2). The most expensive areas are all in central Colombo (Kurunduwatta, Kollupitiya, Bambalapitiya, Havelock Town, Wellawatta). The cheapest are on the outer eastern and south-eastern edge of the district (Kosgama, Bope, Padukka, Meegoda, Dampe).

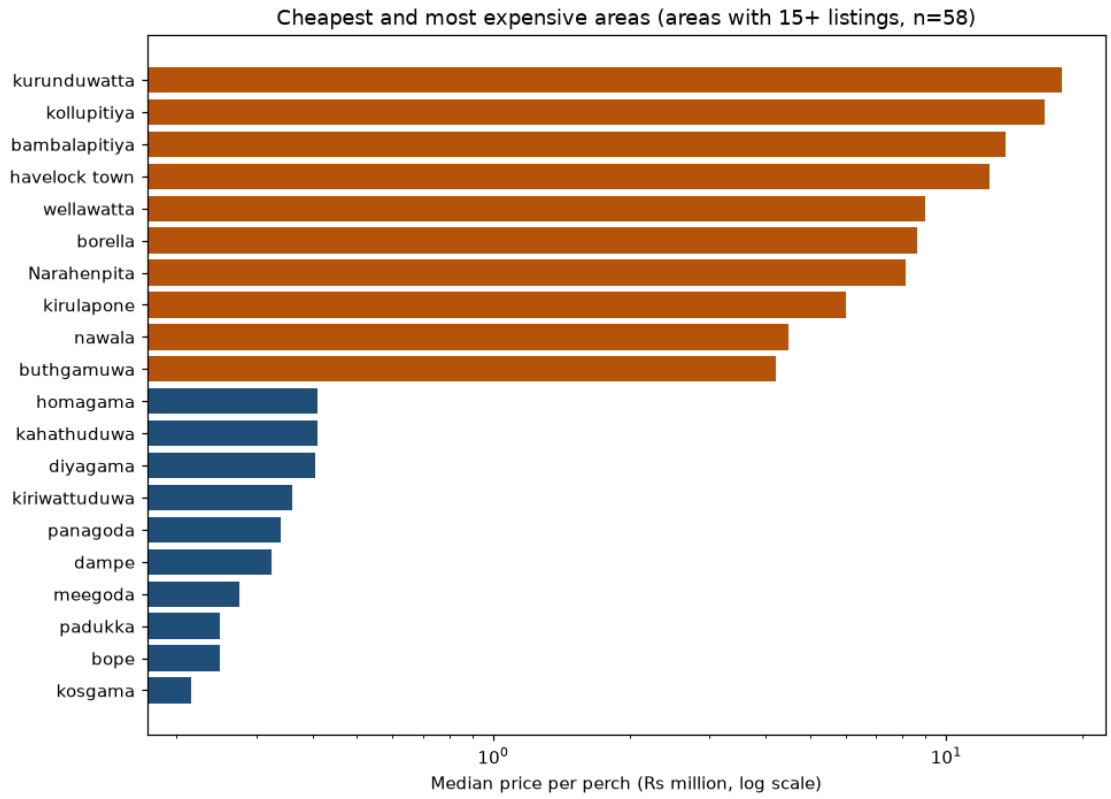


Figure 2: Median price per perch in the 10 cheapest and 10 most expensive areas (log scale)

Two modelling decisions follow from this. First, the area is likely to be the single most informative piece of information. Second, with so many rare areas, one-hot encoding would create many columns that each describe only a handful of listings. Cross-fitted target encoding and CatBoost's native handling cope much better with rare categories (Section 3.6).

Correlations. Spearman correlation was used instead of Pearson, because prices are heavily skewed and the relationships only need to be consistently increasing or decreasing, not straight lines (Figure 3, left).

- **Distance from Fort** has the strongest relationship with price ($\rho = -0.81$), followed by **distance to the nearest grade-A government school** (−0.78).
- The **number of supermarkets** (+0.76) and **banks** (+0.75) within 2 km are the strongest positive relationships. They act as measures of how built-up and commercial an area is.
- **Land size** is positively related to price per perch ($\rho = +0.40$), which is unexpected; Section 6.1 discusses why.
- The **land type** columns have weak relationships with price on their own ($|\rho| < 0.15$).

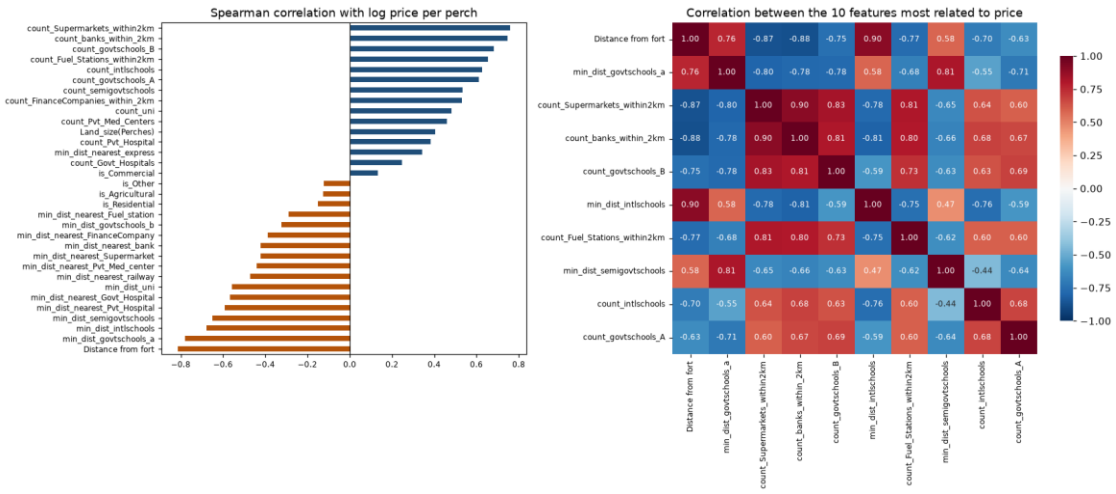


Figure 3: Spearman correlation of each feature with log price (left) and between the ten features most related to price (right)

The right-hand side of Figure 3 shows that the location features are **strongly correlated with each other**. Among the ten features most related to price, **22 of the 45 pairs have |ρ| above 0.7**; for example, distance from Fort and the number of banks within 2 km have ρ = −0.88. These features mostly describe the same thing: how central and built-up an area is. This multicollinearity is why Random Forest was allowed to consider only a subset of features at each split (Section 4.4), and why feature importances must be read for location as a whole rather than feature by feature (Sections 6.1 and 6.4). It was also the first sign of the finding in Section 3.7, that the location features are recorded per area.

The final modelling table has **4,275 rows, 33 features and the log target**.

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4. Methodology

4.1 Data split and validation

- **One listing was set aside** for the inference demonstration (a 9.7-perch plot in Dampe) before anything else. It was not used in training, tuning or testing.
- The remaining 4,274 rows were split **80/20** into training (3,419) and test (855) sets. The split was stratified by price thirds, so cheap and expensive land are equally represented in both sets.
- **Validation** was done with **5-fold cross-validation inside the training set**: each candidate model and hyperparameter setting was trained on four folds and scored on the fifth, five times over. All tuning and model **selection** used these validation scores only. The **test set** was used once per model at the end, so the choice of final model is not biased towards whatever happened to suit the test split.

Set	Size	Used for
Training	3,419 listings	Fitting the models

Set	Size	Used for
Validation	5-fold CV within training	Hyperparameter tuning and model selection
Test	855 listings	Final, one-off performance check
Held-out	1 listing	Inference demonstration

Metrics. The main metric is the **RMSE of the log price**. Because the target is in logs, an RMSE of 0.37 means predictions are typically within a factor of about e^0.37 ≈ 1.45 of the true price. R^2 on the log scale is also reported. Converted back to rupees, the MAE, the mean absolute percentage error (MAPE) and the median absolute percentage error are reported. The **median** percentage error is the most useful single number for a buyer, since it isn't dominated by a few very cheap plots where a small rupee error is a large percentage.

4.2 Models

Model	Why it was included
Ridge regression	Taught. Baseline: if tree models can't beat a linear model clearly, their extra complexity isn't justified
Random Forest	Taught in the module (comparison model). Averages many deep, de-correlated trees; robust to noise
XGBoost	Taught in the module (comparison model). Gradient boosting: each tree corrects the previous ones' errors
LightGBM	Taught. Gradient boosting that grows trees leaf-wise; a check that XGBoost's result isn't implementation-specific
CatBoost	Taught (lecture and MLOps workshop). Built for categorical features such as the 160-category `Address`
Stacking	Taught. Combines the tuned RF, XGBoost and CatBoost with a Ridge meta-model; helps when the base models make different mistakes
NGBoost	**The new algorithm, not covered in the lectures** (Section 4.3). Predicts a probability distribution, giving a price range

Deep learning was not used, following the module guideline, and with 4,000 tabular rows it would not be expected to beat gradient boosting anyway.

4.3 Selection of a new algorithm: NGBoost

The models covered in the lectures were linear, Ridge and Lasso regression, logistic regression, SVM, k-NN, decision trees, bagging, Random Forest, gradient boosting, AdaBoost, XGBoost, LightGBM, CatBoost, voting and stacking ensembles, and Explainable Boosting Machines (plus clustering methods). **NGBoost (Natural Gradient Boosting; Duan et al., 2020)**, from the Stanford ML Group, is not among them. It was chosen as the project's new algorithm (`05_ngboost.ipynb`).

Why it was selected.

1. **Every taught model gives one number, but this problem needs a range.** Asking prices are noisy: similar plots in the same area are advertised at very different prices (Section 3.4), and 88 of the 160 areas have fewer than 10 listings (Section 3.8). A single predicted price hides how uncertain it is. NGBoost predicts a **probability distribution** for each plot, so it can say "80% likely between Rs X and Rs Y". That is more honest and more useful to a buyer or seller.
2. **It suits the data.** It's a boosting method for tabular data and works well with a few thousand rows. It uses the same preprocessing as the other tree models (cross-fitted target encoding of the area).
3. **It fits the guideline's restrictions.** It's a classical, non-deep-learning method.

How it works. NGBoost treats prediction as estimating a distribution. Here it assumes the **log price** of a plot follows a Normal distribution with its own mean μ(x) and standard deviation σ(x), which means the price itself is log-normal. Both μ and σ are learned by boosting:

- At each step, one small regression tree is fitted **per parameter** (one for μ, one for log σ). The trees are fitted to the gradient of a proper scoring rule (the log-likelihood), not to plain residuals.
- It uses the **natural gradient**: the ordinary gradient rescaled by the Fisher information of the distribution. This makes the updates independent of how the distribution is parametrised, which is what makes learning the spread σ stable. Plain gradient boosting on two parameters tends to be unstable.
- A line search chooses the step size, and the learning rate shrinks it, as in other boosting methods.

The result is a full predictive distribution per plot. Its mean is the point prediction, and its quantiles give price ranges ($\mu \pm 1.28\sigma$ for 80%).

How it differs from the models taught in the module.

Model	What it predicts	Main difference from NGBoost
Decision tree / Random Forest / bagging	One number (average of leaves or trees)	No spread estimate; uncertainty isn't modelled

Model	What it predicts	Main difference from NGBoost
Linear / Ridge / Lasso regression	One number; one constant error variance for all plots	NGBoost is non-linear and lets the uncertainty differ from plot to plot
Logistic regression	Probability of a **class**	NGBoost gives a probability distribution over a **continuous** price
k-NN, SVM	One number from neighbours or a kernel	No distribution; need scaling; less suited to the 160-level category
Gradient boosting, AdaBoost, XGBoost, LightGBM, CatBoost	One number: trees fitted to residuals of a loss	NGBoost boosts **two** distribution parameters with the **natural gradient** of the likelihood
Stacking, voting	One number combining other models	—

Tuning. As for every other model, NGBoost was trained first with defaults, then tuned with `RandomizedSearchCV` (20 candidates, 5-fold CV, minimising log RMSE). The selected settings were 489 boosting iterations, learning rate 0.0076, base trees of depth 5 with at least 15 listings per leaf, and no row or column subsampling. Tuning reduced the cross-validated RMSE from 0.437 to **0.428** and raised test R^2 from 0.848 to **0.863**.

How it performed.

- **Point accuracy:** NGBoost ranks **third of all eight models** on cross-validated error (0.4278), only 0.0013 behind the stacked ensemble. Its test R^2 is 0.863 and median error 13.1% (Section 5.1).
- **Price ranges are reliable:** its 80% ranges contain **81.9%** of actual test prices (Section 6.7).
- **Ranges reflect the data:** they are wider for areas with few listings (Section 6.7).
- **Consistent with the final model:** 99.9% of the stacked model's predictions fall inside NGBoost's 80% range.

NGBoost is therefore used **alongside** the stacked ensemble in the application. The ensemble gives the price, because it was marginally best on cross-validation, and NGBoost gives the **likely range** around it (Section 7.2).

Implementation note. The `ngboost` library (0.5.11) is used. A small wrapper (`ngboost_wrapper.py`) tells scikit-learn when the model is fitted. Without it, the scikit-learn 1.9 pipeline check made every cross-validation score come back as `NaN`.

4.4 Hyperparameter design

Every model was first trained with **library defaults**, then tuned with `RandomizedSearchCV` (5-fold CV, minimising log RMSE). The search ranges were chosen for this dataset of about 3,400 training rows and 33 features, whose prices are noisy because they are asking prices, not sale prices.

Random Forest (25 candidates)

Parameter	Search range	Selected	Reasoning
`n_estimators`	200-800	271	More trees reduce variance, but the gain flattens after a few hundred
`max_depth`	10, 15, 20, 30, unlimited	20	Unlimited trees can memorise individual listings
`min_samples_leaf`	1, 2, 3, 5, 8	3	Leaves of a single listing would fit noise in the asking prices
`max_features`	0.3-1.0	0.3	Many features are highly correlated (Section 3.7); considering fewer at each split makes the trees more different from each other
`max_samples`	70%, 85%, 100%	100%	Bootstrap sample size

XGBoost (40 candidates)

Parameter	Search range	Selected	Reasoning
`learning_rate`	0.01-0.2 (log scale)	0.0127	Small steps generalise better but need more trees
`n_estimators`	300-1500	1066	Paired with the small learning rate
`max_depth`	3-8	5	Boosted trees should be shallow; each corrects only part of the error
`min_child_weight`	1-10	1	Minimum data

Parameter	Search range	Selected	Reasoning
			needed in a leaf
`subsample` / `colsample_bytree`	0.6-1.0 / 0.5-1.0	0.79 / 0.78	Random row and column sampling to reduce overfitting
`reg_lambda` / `gamma`	0.1-10 / 0-0.3	4.04 / 0.18	L2 penalty on leaf values, and minimum gain required to split

LightGBM (30 candidates) selected `num_leaves` 15, `learning_rate` 0.0125, `n_estimators` 820, `min_child_samples` 11, `subsample` 0.60, `colsample_bytree` 0.90 and `reg_lambda` 0.25.

CatBoost (12 candidates from a grid of depth 4/6/8, learning rate 0.03/0.06/0.1, `l2_leaf_reg` 1/3/5/10 and 800/1200 iterations) selected depth 8, learning rate 0.03, `l2_leaf_reg` 5 and 800 iterations.

A pattern runs through all of these choices: **every model chose strong regularisation**. Random Forest considers only 30% of features per split and needs 3 listings per leaf. XGBoost and LightGBM chose very small learning rates, sub-sampling and small trees (LightGBM's 15 leaves is near the bottom of its range). This is consistent with what Section 3.7 found: within an area there is little real signal beyond size and type, so the best models are the ones that avoid chasing noise.

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5. Results

5.1 Model comparison

Model	CV RMSE (log)	Test RMSE (log)	Test R^2 (log)	Test MAE (Rs/perch)	Test MAPE	Test median APE
Stacking (RF + XGB + Cat)	**0.4265**	**0.3677**	**0.866**	554,620	23.2%	11.8%
LightGBM (tuned)	0.4274	0.3726	0.862	567,464	24.5%	13.5%
NGBoost (tuned, new algorithm)	0.4278	0.3714	0.863	572,292	24.8%	13.1%
XGBoost (tuned)	0.4282	0.3729	0.862	561,203	24.2%	12.2%

Model	CV RMSE (log)	Test RMSE (log)	Test R^2 (log)	Test MAE (Rs/perch)	Test MAPE	Test median APE
Random Forest (tuned)	0.4289	0.3742	0.861	552,220	23.1%	11.5%
CatBoost (tuned)	0.4311	0.3677	0.866	565,099	24.0%	12.3%
NGBoost (default)	0.4372	0.3921	0.848	596,068	26.4%	14.5%
Random Forest (default)	0.4450	0.4092	0.834	537,807	24.3%	10.3%
XGBoost (default)	0.4490	0.4138	0.830	594,944	25.6%	11.0%
Ridge (baseline)	0.4509	0.4063	0.836	594,270	28.8%	16.5%

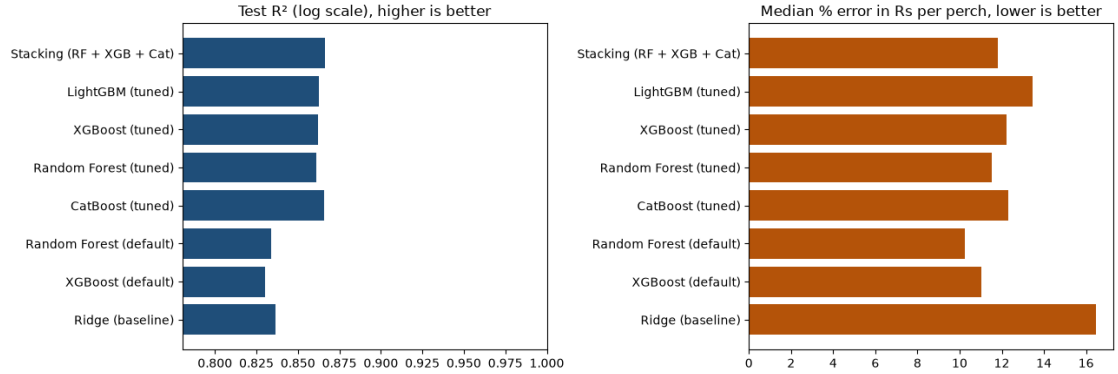


Figure 4: Test R^2 and median percentage error for each model (notebook 02 models; NGBoost is in the table above)

The main observations are:

1. **Tuning clearly helped.** Tuning reduced the cross-validated RMSE of Random Forest from 0.445 to 0.429 and of XGBoost from 0.449 to 0.428. Test R^2 rose from about 0.83 to 0.86 for both. The defaults (unlimited tree depth for Random Forest, a fast learning rate for XGBoost) overfit this noisy data.
2. **The tuned tree models are very close to each other.** All six tuned models, including NGBoost, fall within 0.005 RMSE in cross-validation, which is smaller than the variation between folds. No single algorithm is clearly better here; the dataset, not the algorithm, limits the accuracy.
3. **The linear baseline is already strong** (R^2 0.836). The tree models improve on it by about 3 points of R^2 and cut its median error from 16.5% to about 12%. That

improvement is real and useful, but modest. Section 6.3 explains why: most of the price is explained by the area, which even a linear model captures through the target-encoded area feature.

4. **The default Random Forest has the lowest median error (10.3%)** despite having a worse RMSE. RMSE penalises large errors heavily, so tuning for RMSE made the models more careful about the expensive outliers, at a small cost on typical listings. Which one is "better" depends on whether large errors or typical errors matter more. For a pricing tool, avoiding large mistakes is the better priority, so the RMSE-based selection was kept.

5.2 Final model

The **stacked ensemble** had the lowest cross-validated RMSE (0.4265) and was selected as the final model. On the test set it also shares the best R^2 (0.866) with CatBoost. The Ridge meta-model gave weights of **0.47 to Random Forest, 0.30 to CatBoost and 0.22 to XGBoost**. The ensemble leans most on Random Forest, whose averaging approach makes different errors from the two boosting models.

NGBoost, the new algorithm, is almost as accurate (third on cross-validation), and it is the only model that also gives a price **range**. The deployed application therefore uses the stacked ensemble for the price and NGBoost for the range around it (Sections 6.7 and 7.2).

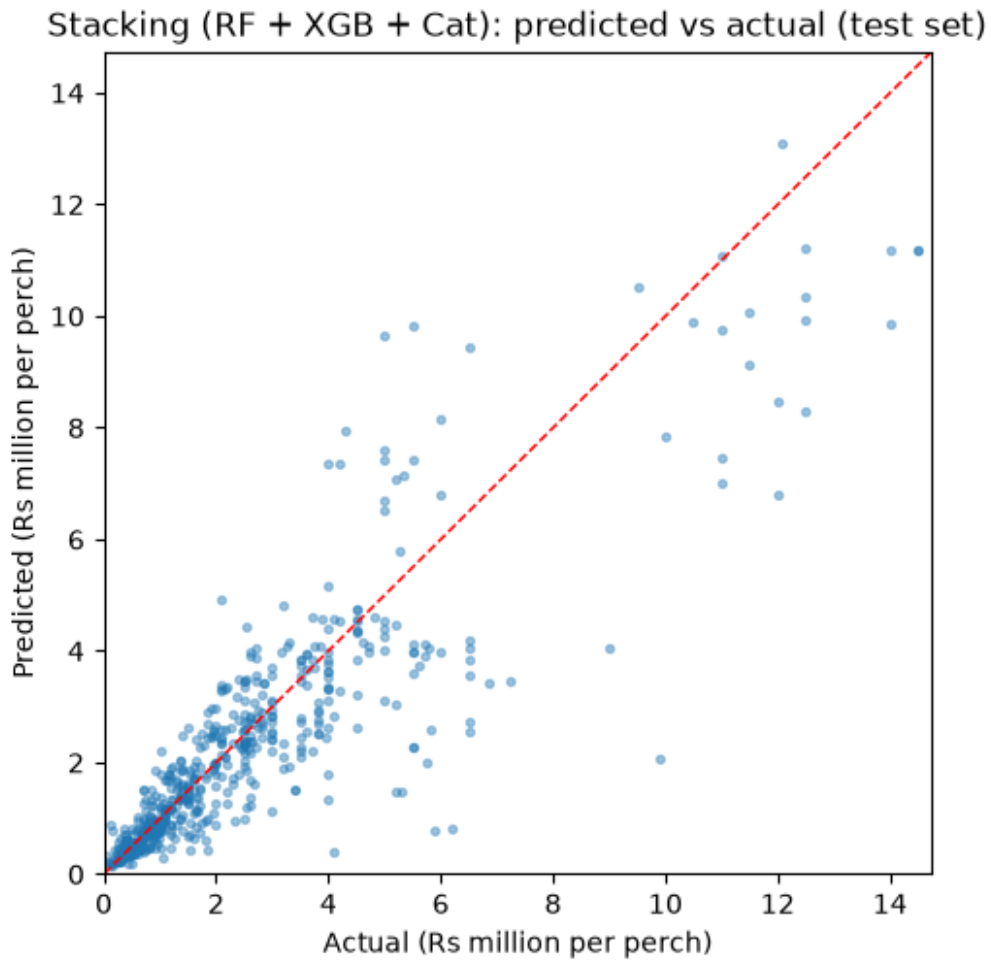


Figure 5: Predicted vs. actual price per perch on the test set (final model)

Figure 5 shows the predictions follow the actual prices well across the range. Two patterns are visible: the spread grows with price (expected, since errors are roughly proportional), and the very highest prices are under-predicted. Rare, very expensive plots in central Colombo are pulled towards the average of their area.

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6. Interpretation

6.1 Which features matter

Permutation importance was used: each feature is shuffled in the test set, and the importance is how much the RMSE increases. Unlike the built-in importances, this works the same way for every model, including the stacked ensemble.

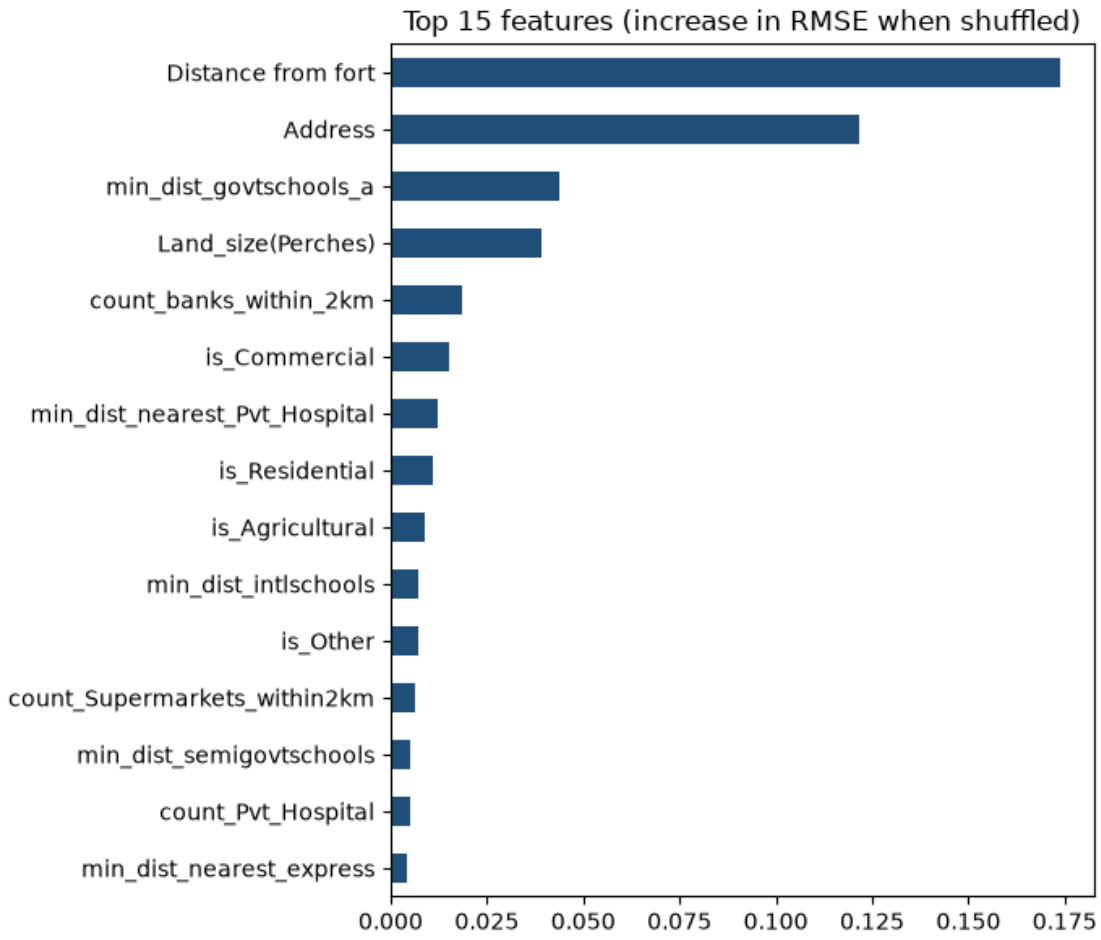


Figure 6: Permutation importance of the final model

The two strongest features are **distance from Colombo Fort** and **area**. Because all location features are constant within an area (Section 3.7), these two carry largely the same information, and the importance is divided between them somewhat arbitrarily. The fair reading is that **location as a whole dominates**. Among the remaining features, **land size** and the **land type** columns (commercial, residential, agricultural) matter most, since they are the only features that vary between plots in the same area.

The direction of the main effects, measured directly from the data (median price per perch by quartile):

Feature	Lowest quartile	Highest quartile	Direction
Distance from Fort	Rs 3.75 M (3-13 km)	Rs 0.44 M (22-43 km)	About **8.5× cheaper** furthest from the city
Distance to railway	Rs 2.60 M (< 1 km)	Rs 0.44 M (> 6.7 km)	Closer to rail = more expensive
Banks within 2 km	Rs 0.59 M (0-2 banks)	Rs 3.70 M (15+ banks)	Proxy for how commercial/urban

Feature	Lowest quartile	Highest quartile	Direction
			the area is
Land size	Rs 0.65 M (≤ 7 perches)	Rs 2.35 M (> 16 perches)	Larger plots cost **more** per perch

The last row is counter-intuitive, since bulk land is usually cheaper per unit. It is most likely explained by **where** the plots are, not by their size. Small plots of 6-10 perches are typical of suburban housing developments sold off a larger block, while large plots in the dataset are more often commercial land in built-up areas. This is a reminder that these figures show associations, not causes.

6.2 Price-band classification

For the classification extension, the training prices were split into thirds: **Low** (below Rs 672,486 per perch), **Medium** (up to Rs 1,800,000) and **High** (above). The band edges were calculated from the training data only.

Model	Accuracy	Macro F1
Random Forest classifier (tuned)	**0.905**	**0.905**
XGBoost classifier (tuned)	0.897	0.896
Final regression model, prediction converted to a band	0.892	0.892

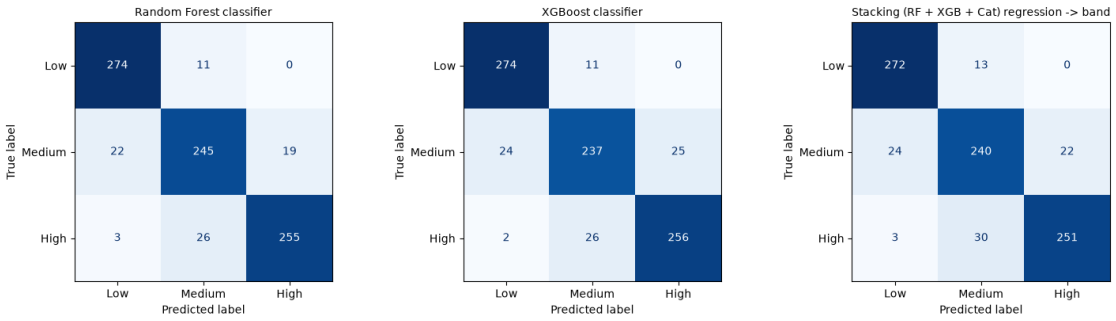


Figure 7: Confusion matrices for the three approaches

All three approaches reach about **90% accuracy**, and almost every mistake is between **neighbouring** bands. A Low plot is almost never classified as High or the other way round (Figure 7). **Medium** is the hardest class (F1 0.86, against 0.94 for Low and 0.91 for High), as expected: plots near either edge of the Medium band can fall on either side through small price differences. Training a dedicated classifier gave only slightly better results than converting the regression output to a band. This shows the two framings capture the same underlying information.

6.3 Seen areas vs. unseen areas

Because the location features are fixed per area (Section 3.7), the price variation can be split into two parts. **81% of the variance in log price is between areas** and only **19% is within areas**. Most of what the models learn is therefore "what does land cost in this area", and the remaining 19% has to be explained by land size, land type and noise.

The random split used in Section 5 tests **seen areas**: almost every area in the test set also appears in the training set. That is the right test for the application, which only offers areas from the dataset. It doesn't show whether the models have learnt anything **general** about location. To check that, the models were also evaluated with **grouped cross-validation** (`GroupKFold`, grouped by area). Each fold's test areas are completely absent from its training data, so the model can't use the area name and has to rely on the 27 location features (`03_unseen_areas.ipynb`).

Model	CV RMSE, seen areas	CV R^2, seen areas	CV RMSE, unseen areas	CV R^2, unseen areas
Ridge (baseline)	0.451	0.808	0.672	0.541
Random Forest (tuned)	0.429	0.827	0.557	0.688
XGBoost (tuned)	0.428	0.827	0.576	0.665
CatBoost (tuned)	0.431	0.825	**0.542**	**0.703**

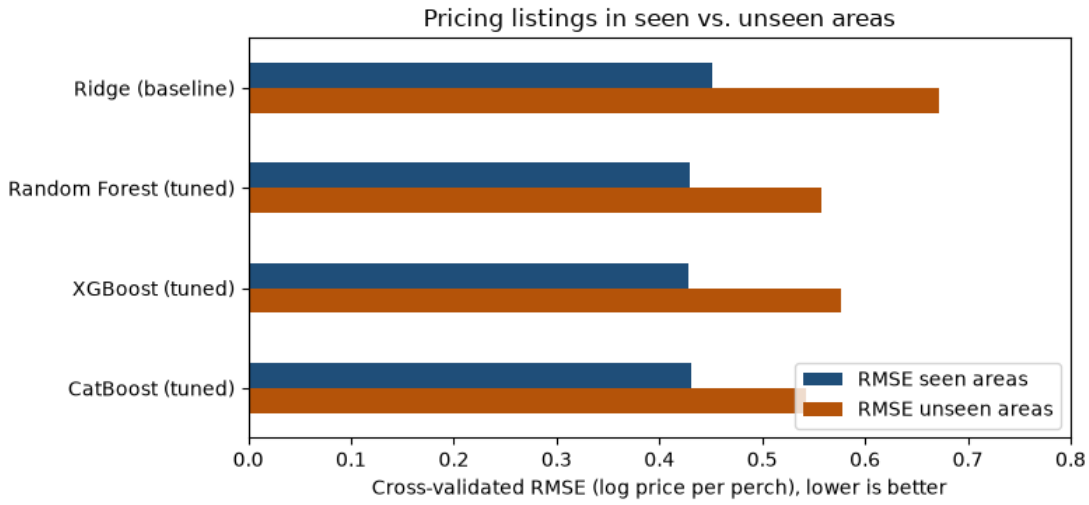


Figure 8: Cross-validated error for seen vs. unseen areas

(These are cross-validation figures on the training set, which is why the seen-area R^2 of about 0.83 is a little lower than the 0.86 measured on the test set in Section 5.)

Three conclusions follow:

1. **Unseen areas are much harder.** The error rises from about 0.43 to 0.54-0.58 for the tree models. In practical terms, a typical prediction moves from within a factor of about 1.5 of the true price to within a factor of about 1.7. The models are noticeably less reliable for an area outside the dataset, which is why the application only offers known areas.
2. **This is where the tree models really earn their place.** With seen areas, the tree models beat Ridge by only about 0.02 in RMSE. With unseen areas the gap grows to over 0.1: R^2 0.69-0.70 against 0.54. Once the area name isn't available, price has to be inferred from the distances and counts, and those relationships are non-linear (the effect of distance to Fort flattens out beyond about 20 km, for example). Trees capture that; a linear model can't.
3. **The location features do carry real, general information.** An R^2 of 0.70 for areas the model has never seen means that distance to the city, transport and the density of banks, schools and hospitals explain most of the difference in price **between** areas. Within an area, as noted, they can't help, because they don't vary.

CatBoost generalises best to new areas. Its ordered target statistics treat an unknown area cleanly as "no information", while the stacked ensemble's target encoder falls back to the overall average. For a future version of the application that allowed **new** areas, CatBoost on its own would be the better choice. For the current application, restricted to known areas, the stacked ensemble remains the best model.

6.4 Explainability with SHAP and LIME

Permutation importance (Section 6.1) shows *which* features matter overall. It doesn't show in which **direction** they push the price, or why the model priced one particular plot as it did. Two explainability methods were applied to the **final stacked model** for this (`04_explainability.ipynb`):

- **SHAP** splits every prediction into a contribution from each feature, so that the baseline plus the sum of the contributions equals the prediction. Because a stacked ensemble can't be opened up by tree-specific methods, the model-agnostic `PermutationExplainer` was used, with 50 training listings as the background, to explain 120 test listings.
- **LIME** explains a single prediction. It creates 5,000 perturbed copies of the listing, asks the model to price each one, and fits a simple weighted linear model to the answers.

Both methods work on numbers, so each area name was given an integer code, and a small wrapper converts the codes back to names before calling the model. A check confirmed that the wrapper gives exactly the same predictions as the model. All explanations are on the **log** scale, where a contribution of −0.10 means roughly 10% lower in price.

Global explanation (SHAP).

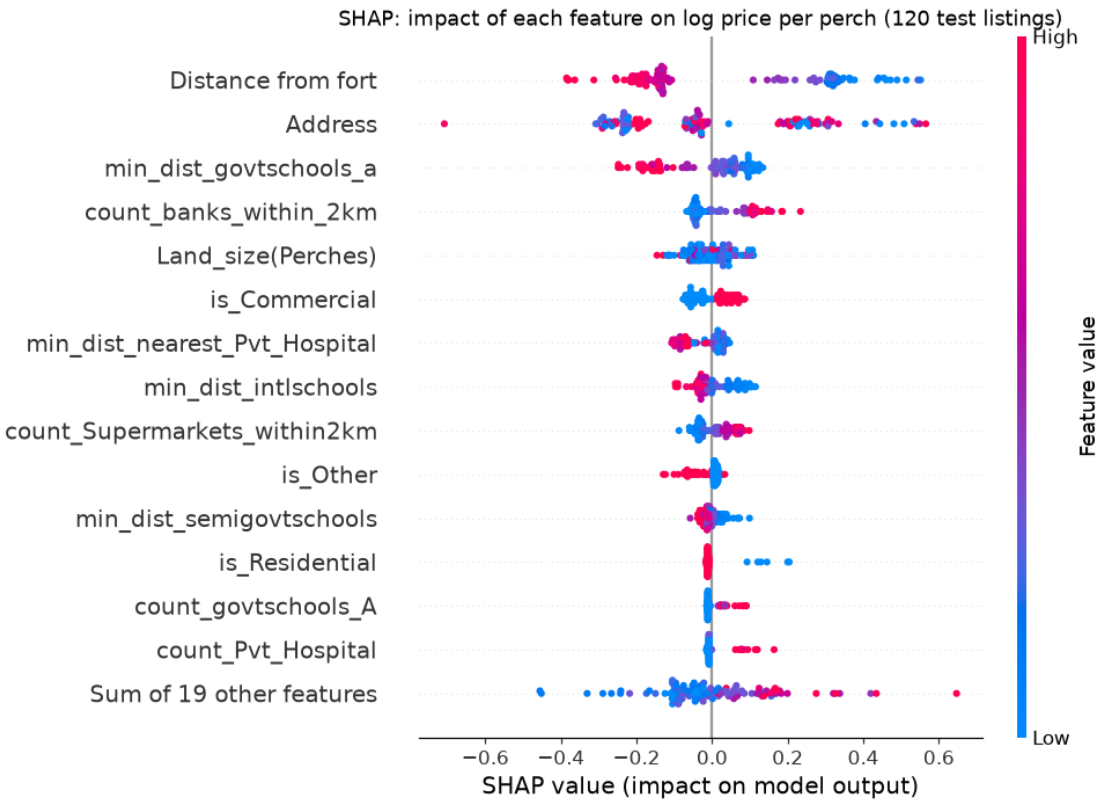


Figure 9: SHAP summary of 120 test listings (red = high feature value, blue = low)

Figure 9 confirms the direction of the main effects found in Section 6.1, now for the model itself rather than the raw data:

- **Distance from Fort:** a long distance (red) pushes the price down by up to about 0.4 (around 33% lower). Short distances (blue) push it up by as much as 0.5.
- **Banks within 2 km** and **supermarkets within 2 km:** more of them (red) raise the price, a sign of a built-up, commercial area.
- **Distance to grade-A government schools** and **private hospitals:** being far from them lowers the price.
- **Land type:** commercial land (`is_Commercial` = 1) is priced higher, and land listed as "Other" lower.
- **Land size** has small effects in both directions, which matches its low importance.

Grouped view. Section 3.7 showed that the location features are constant within an area, so SHAP splits the credit between them somewhat arbitrarily. Adding the contributions together gives a fairer summary:

Feature group	Share of total SHAP impact
Location (area + 27 distance and count features)	**87.1%**
Land type (4 columns)	8.6%

Feature group	Share of total SHAP impact
Land size	4.3%

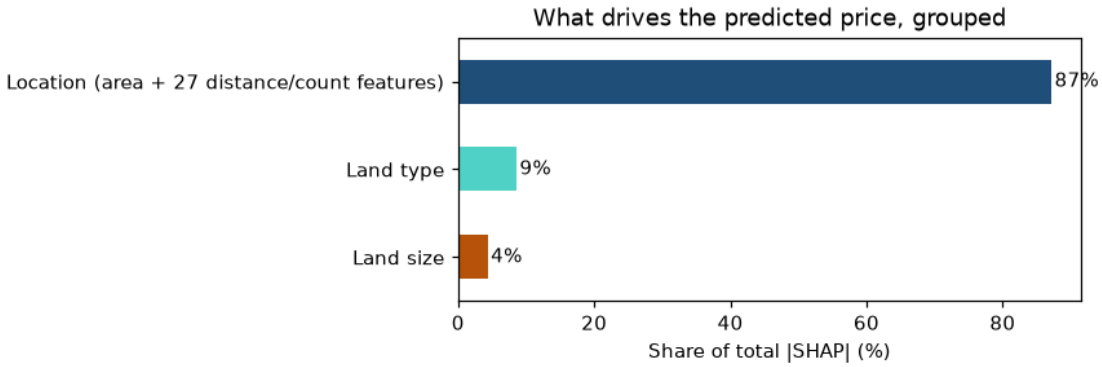


Figure 10: SHAP impact grouped into location, land type and land size

This matches the finding in Section 6.3 that 81% of price variation lies between areas, reached in a completely different way. **Where the land is explains almost all of the model's decisions.**

Local explanation of the held-out listing (SHAP). The Dampe listing kept aside for inference (Section 7.1) had an actual price of Rs 300,000 per perch and a predicted price of Rs 332,620.

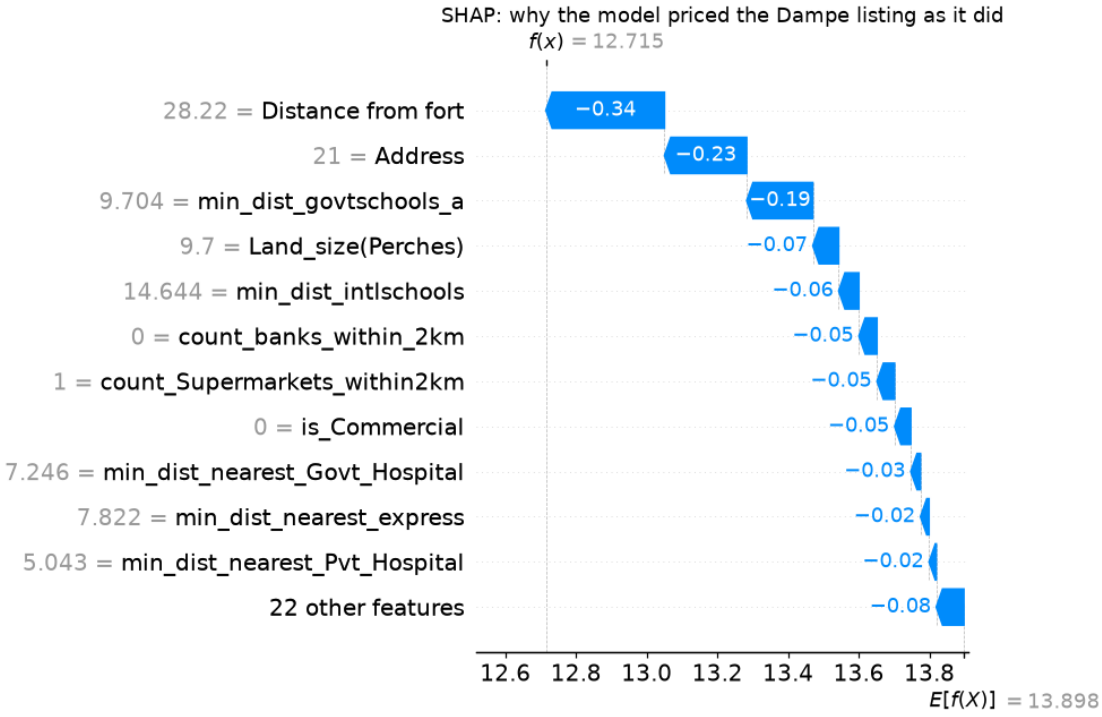


Figure 11: SHAP waterfall for the held-out Dampe listing (Address is shown as its area code; 21 = Dampe)

The model starts from the average prediction (13.898 on the log scale, about Rs 1.09 million per perch). Every major feature then pulls the price **down**. The plot is 28 km from Fort (−0.34), Dampe is a low-priced area (−0.23), the nearest grade-A school is almost 10 km away (−0.19), and there are no banks and only one supermarket within 2 km. The contributions add up to −1.183, which gives 12.715 on the log scale: exactly the model's prediction of Rs 332,620. In plain terms, the model sees an outer-suburban, non-commercial plot far from city amenities and prices it at about a third of the average.

Local explanation of the same listing (LIME).

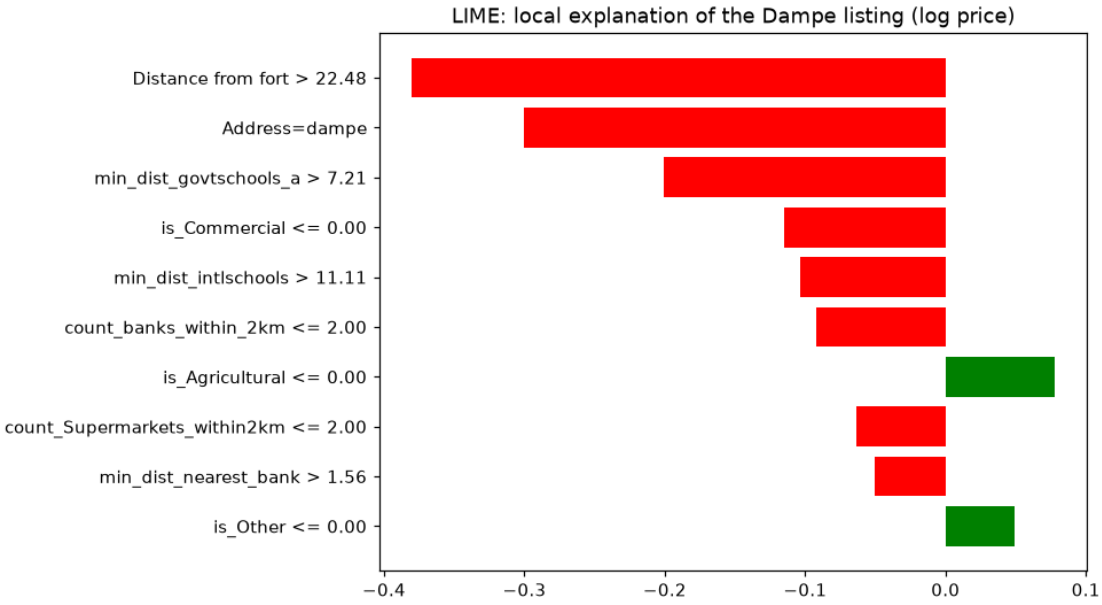


Figure 12: LIME explanation for the held-out Dampe listing (red = lowers the price, green = raises it)

Comparison of the two methods for the Dampe listing:

Feature	SHAP contribution	LIME weight	Same direction?
Distance from Fort	−0.336	−0.380	Yes
Address (Dampe)	−0.233	−0.300	Yes
Distance to grade-A govt. school	−0.187	−0.201	Yes
Land size	−0.073	–	Not in LIME's top 10
Distance to international school	−0.057	−0.104	Yes
Banks within 2 km	−0.051	−0.092	Yes
Supermarkets within 2 km	−0.050	−0.063	Yes
Commercial land	−0.046	−0.115	Yes

The two methods **agree on the top three drivers, in the same order and with similar sizes**, and on the direction of every feature they both list. This agreement makes the explanation more trustworthy, since the methods work in very different ways.

Limits of the explanations.

- LIME's local linear model has an R^2 of only **0.29**, so it captures the model's behaviour around this listing only roughly. It is best read as a check on direction, with SHAP as the main explanation.
- Both methods change one feature at a time. Here the location features always move together, so some perturbed listings combine one area's name with another area's distances, a combination that never occurs in the data. This is why the grouped SHAP view is the most reliable reading, and why individual location features should be read as a group.

6.5 What CatBoost learned, and partial dependence

CatBoost SHAP. Because CatBoost is a tree model, its SHAP values can be computed **exactly** with TreeSHAP, using CatBoost's built-in calculation. No sampling is needed, unlike the stacked model. The contributions plus the baseline reproduce CatBoost's predictions exactly.

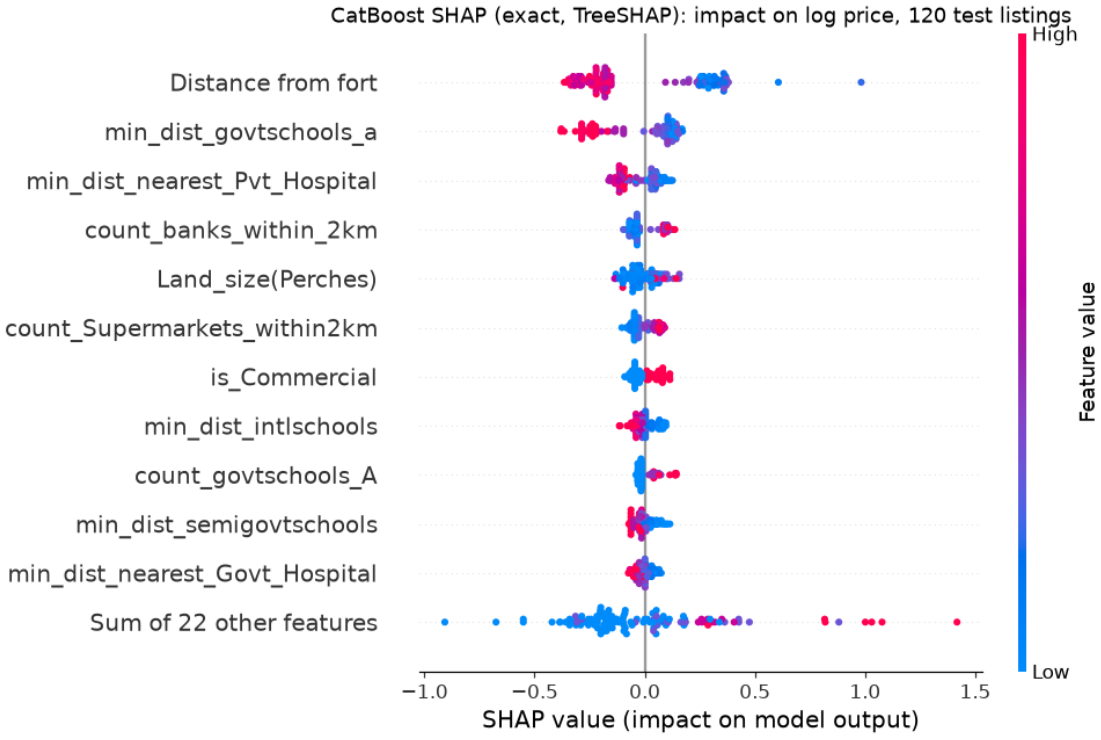


Figure 13: Exact SHAP values of CatBoost for 120 test listings (red = high feature value, blue = low)

Feature group	CatBoost (exact SHAP)	Final stacked model
Location (area + 27 distance/count features)	**86.5%**	87.1%
Land type	9.0%	8.6%
Land size	4.6%	4.3%

CatBoost and the final model agree almost exactly on what drives prices. The interesting difference is in **how** CatBoost uses location. Its top features are distance from Fort, distance to a grade-A government school, distance to a private hospital and the number of banks within 2 km. The stacked model relies more on the area name. Because CatBoost's encoding of `Address` is deliberately conservative (ordered target statistics), it learns location through the interpretable distance features. This is also why it generalises best to unseen areas.

Partial dependence. A partial dependence plot (PDP; Friedman, 2001) shows the model's average prediction as one feature is varied, with the other features left as they are in the data. It was computed on 300 test listings.

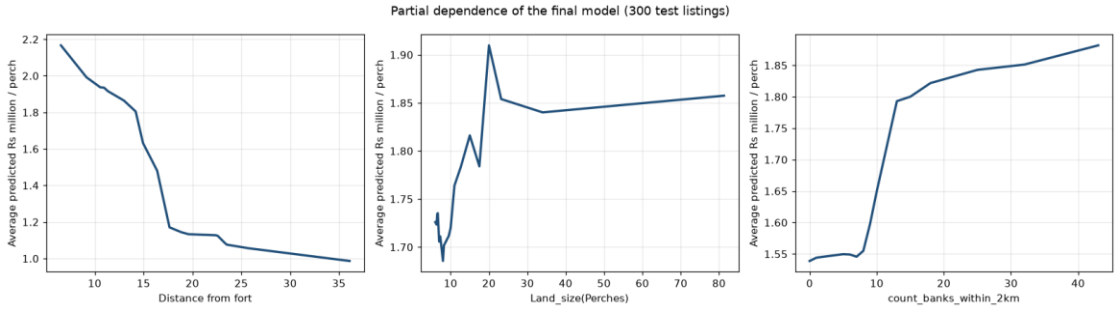


Figure 14: Partial dependence of the final model on distance from Fort, land size and banks within 2 km

- **Distance from Fort:** going from 6.5 km to 36 km roughly **halves** the average predicted price, from about Rs 2.17 million to Rs 0.99 million per perch. The steepest fall is between about 14 and 18 km, roughly where the city gives way to the outer suburbs.
- **Banks within 2 km:** the price rises from about Rs 1.54 million (no banks) to Rs 1.88 million (43 banks), mostly between 8 and 13 banks. This is the threshold between a residential suburb and a commercial town centre.
- **Land size:** almost flat (about +8% from 6 to 80 perches). Once location is accounted for, plot size has little effect on the price per perch.

As with SHAP and LIME, location features never vary on their own within an area, so these curves show the **direction** the model has learned rather than exact causal effects.

6.6 Does the model agree with domain knowledge?

What the model learned	Domain knowledge (Colombo land market)	Agrees?
Price falls steeply with distance from Colombo Fort	Colombo's central business district and Colombo 1-7 have the highest land values; prices fall towards the suburbs	**Yes**
More banks and supermarkets nearby means higher prices	Commercial town centres (e.g. Nugegoda, Maharagama) command a premium	**Yes**
Closeness to grade-A government schools raises prices	Demand for popular school catchment areas is a well-known driver of Sri Lankan property prices	**Yes**
Commercial land is priced higher; "Other" land lower	Commercial plots earn more; unusual plot types sell at a discount	**Yes**
Larger plots cost more *per perch* (raw data)	Bulk land is usually *cheaper* per perch	**No, explained:** the PDP shows almost no size effect once location is held fixed. The raw pattern comes from large plots being more common in built-up areas

Overall, the model has learned the patterns a local estate agent would expect. The one counter-intuitive pattern in the raw data turns out to come from location, not from plot size.

6.7 Uncertainty: NGBoost price ranges

For every test listing, NGBoost gives a mean μ and a standard deviation σ for the log price. An 80% range is μ ± 1.28σ, and a 95% range μ ± 1.96σ, converted back to rupees. A useful range must be **calibrated**: about 80% of actual prices should fall inside the 80% ranges.

Nominal range	50%	60%	70%	**80%**	90%	95%
Actual test prices inside	58.2%	68.1%	76.0%	**81.9%**	88.4%	91.6%

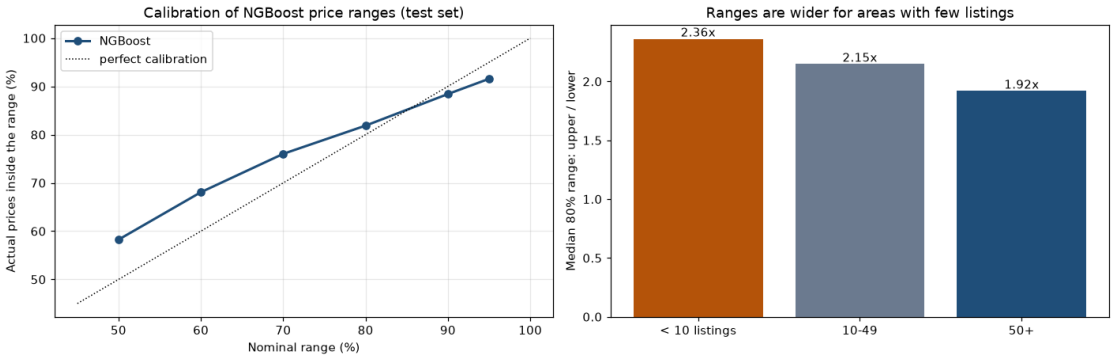


Figure 15: Calibration of NGBoost's price ranges (left) and range width by how many listings an area has (right)

- **The 80% range is well calibrated** (81.9%), which is why the application uses it.
- **The middle ranges are slightly too wide and the 95% range slightly too narrow.** This means the real errors have heavier tails than a Normal distribution: most listings are predicted well, but a few are far more or less expensive than predicted, often unusual plots or likely data-entry errors. A heavier-tailed distribution (e.g. Student-t) could improve this.
- **Ranges are wide but honest.** The median 80% range spans about 2 times from lower to upper bound, for example Rs 1.5M-4.0M per perch for a residential plot in Battaramulla. This reflects how noisy asking prices are, and that location is known only per area. A single price would hide this.
- **Less data means wider ranges.** The median 80% range spans 2.36 times for areas with fewer than 10 training listings, against 2.15 times for 10-49 listings and 1.92 times for 50 or more. The model is appropriately less certain where it has seen less data, which matters for the fairness concerns in Section 8.3.
- **It agrees with the final model:** 99.9% of the stacked model's predictions fall inside NGBoost's 80% range.

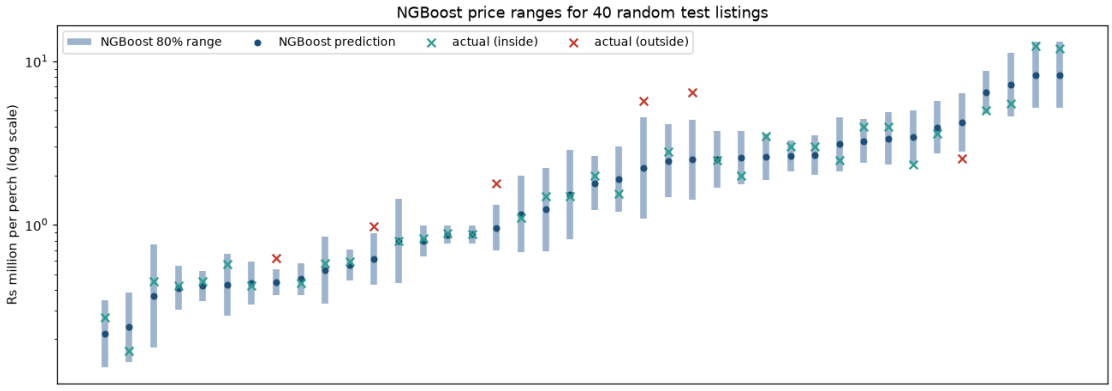


Figure 16: NGBoost predictions and 80% ranges for 40 random test listings; green crosses are actual prices inside the range, red outside

Held-out listing. For the Dampe plot set aside at the start, NGBoost predicts Rs 358,370 per perch, with an 80% range of **Rs 274,480 - Rs 467,899**. The actual asking price, Rs 300,000, is inside the range.

What NGBoost learned (SHAP). SHAP's TreeExplainer supports NGBoost directly, explaining the predicted mean. Location accounts for **86.1%** of its decisions, land type 9.5% and land size 4.4%. This matches the stacked model (87.1%) and CatBoost (86.5%). The area itself and distance from Colombo Fort are its top two features.

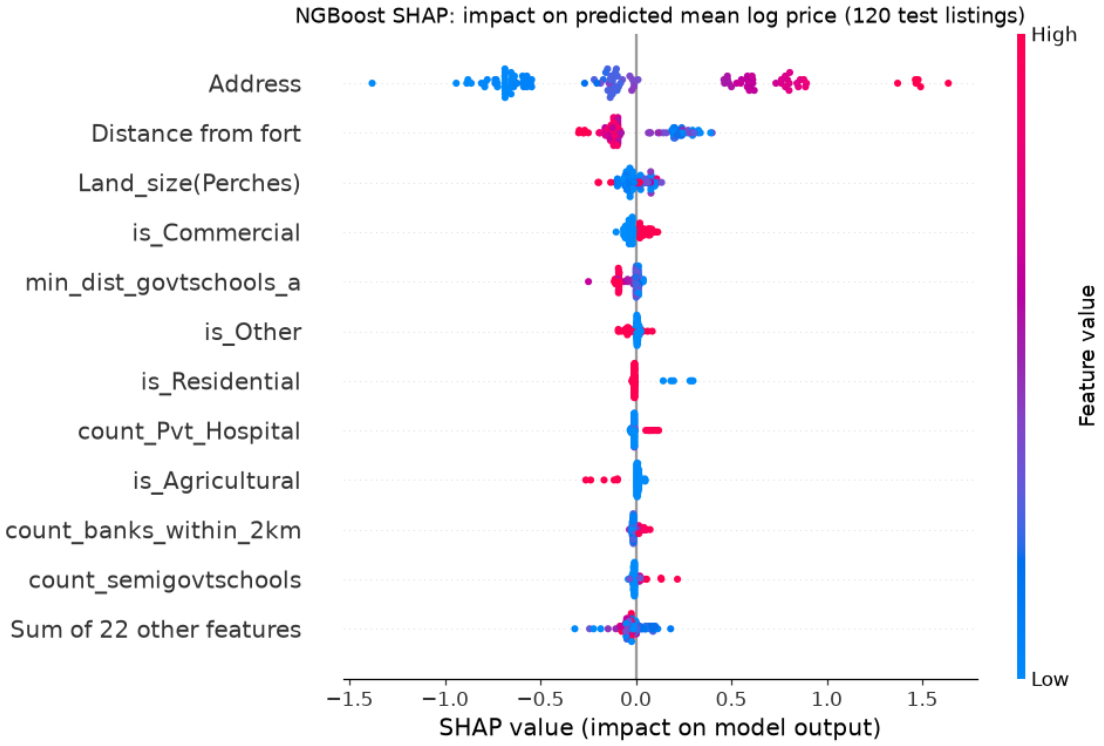


Figure 17: SHAP values of NGBoost's predicted mean for 120 test listings

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7. Inference and Application

7.1 Inference on a held-out listing

The listing set aside at the very start (Section 4.1) was used to demonstrate inference. The model was loaded back from the pickle file, exactly as the application does:

Listing	Dampe, 9.7 perches
Actual asking price	**Rs 300,000** per perch
Predicted price	**Rs 332,620** per perch (**+10.9%**)
Predicted total price	Rs 3,226,419
Predicted band	**Low** (correct)

The error of about 11% is in line with the test set's median error of 11.8%.

7.2 Front-end application

A small web application (`app.py`, built with Streamlit) puts the model in front of a user. The user:

1. selects the **area** from the 160 areas in the dataset,
2. enters the **land size** in perches, and
3. selects one or more **land types**.

The application then shows the **predicted price per perch**, the **estimated total price** and the **price band**. The model, the feature list, the band edges and each area's location values are all loaded from `land_price_model.pkl`. The location features are filled in automatically from the selected area and shown for information only. Section 3.7 showed they are fixed per area, so letting the user edit them would create combinations the model never saw.

Explanations in the application ("Why this price?"). After each prediction, the application explains it with SHAP (`explain.py`), so users see *why* a price was suggested, not just the number:

- **A plain-language sentence** comparing the plot with a typical plot. For example, for Battaramulla: *"Compared with a typical plot (about Rs 1,357,710 per perch), this plot's area and location raises the price by about 104%, its land type lowers the price by about 5%, and its size lowers the price by about 2%."*
- **A bar chart** of those three effects (area & location, land type, land size).
- **A table of the top individual factors.** For Battaramulla these are distance from Colombo Fort (+26%), the area itself (+22%) and the number of banks within 2 km (+7%).

The explanation uses the same model-agnostic SHAP method as Section 6.4, applied to the final stacked model. The "typical plot" baseline is the average prediction over a typical plot in 40 areas. Because the model predicts log price, each contribution is shown as a percentage effect on the price. An explanation takes about half a second. The SHAP values add up exactly to the prediction, and an automated test checks this.

Price range ("Likely range (80%)"). Under the price, the application shows NGBoost's 80% range, with a note that 82% of actual test prices fell inside their 80% range. This tells the user how much to trust the single number.

Example predictions for a 10-perch residential plot:

Area	Predicted price per perch	Likely range (80%, NGBoost)	Band	Main reason (from the explanation)

Area	Predicted price per perch	Likely range (80%, NGBoost)	Band	Main reason (from the explanation)
Kollupitiya (central Colombo)	about Rs 13.1 million	Rs 8.1M - 18.5M	High	Central location close to Fort
Battaramulla (11 km from Fort)	about Rs 2.58 million	Rs 1.5M - 4.0M	High	Location raises the price by about 104%
Kaduwela (18 km from Fort)	about Rs 460,000	Rs 239,000 - 874,000	Low	Location lowers the price by about 63%

To run the application locally: `pip install -r requirements.txt`, then `streamlit run app.py` from the `Assignment_2` folder. It is also deployed online (Section 7.3).

7.3 Repository, deployment and MLOps

The project is published in a private GitHub repository (github.com/amalr-12/colombo-land-price-ml) and the application is deployed on Streamlit Community Cloud:

Live application: https://colombo-land-price-ml-5jekcfuzve5ezgfgjjpqoh.streamlit.app

The deployed app gives exactly the same predictions as the local model (for example, a 10-perch residential plot in Udahamulla: Rs 1,999,440 per perch, High band, both locally and online).

To make the model maintainable rather than a one-off notebook result, a small MLOps pipeline was added:

```
raw Excel -> src/prepare.py -> clean data -> src/train.py -> candidate model
+ metrics.json
                                                                    |
                                        src/promote.py (quality gate) <-+
                                R2 >= 0.85 and not worse than deployed by > 0.005?
                                  yes -> replace land_price_model.pkl (old one back
ed up)
                                  no  -> reject; deployed model unchanged
```

- **Reproducible retraining.** `run_pipeline.py` rebuilds the model from the raw Excel file with the tuned hyperparameters stored in `config/params.json`, using the same seed, held-out listing and split as the notebooks. It took 4.4 minutes and reproduced the notebook: identical cleaned data, and test R^2 0.8659 against 0.8661.
- **Quality gate.** A retrained model replaces the deployed one only if its test R^2 is at least 0.85 and not more than 0.005 below the current model, and NGBoost's 80%

ranges cover 70-90% of test prices. In testing, the gate accepted the real retrained model and rejected candidates with R^2 0.80 and 0.858.

- **Automated tests.** 17 tests check the data rules (row count, no duplicates, no leaky columns), the model's behaviour (the held-out listing reproduces the notebook's prediction; central Colombo is priced above the suburbs; all predictions are finite; explanations add up to the prediction; NGBoost's 80% ranges are ordered and cover 70-90% of test prices) and that the app starts and returns a prediction with its explanation and range.
- **Continuous integration.** A GitHub Actions workflow runs the tests on Linux on every push. The first run passed, confirming that the model trained on Windows loads and predicts correctly with the pinned library versions used by the cloud deployment.
- **Retraining on demand.** A second workflow, started manually from GitHub's Actions tab, runs the full pipeline in the cloud and uploads the candidate model and its metrics for review.
- **Continuous deployment.** Streamlit Community Cloud redeploys the app automatically whenever the `main` branch changes.
- **Pinned dependencies.** `requirements.txt` fixes the exact library versions, because a pickled model must be loaded with the versions it was trained with. pandas is pinned to 2.3.3, because NGBoost depends on `lifelines`, which doesn't yet support pandas 3. The tests confirmed the saved models load and predict identically.

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8. Critical Discussion

8.1 Limitations of the model

- **Location is recorded per area, not per plot** (Section 3.7). This is the biggest limitation. Two plots in the same area, one on a main road and one down a narrow lane, look identical to the model. As a result, accuracy is limited by the data rather than the algorithm: all tuned models score within 0.005 of each other.
- **Weaker on unknown areas.** For areas not in the training data, R^2 falls from about 0.83 to 0.70 (Section 6.3), so the application only offers known areas.
- **Typical error of about 12%**, with larger errors for the rare, very expensive plots in central Colombo, which are pulled towards their area's average (Section 5.2).
- **Wide uncertainty.** NGBoost's 80% ranges typically span about 2 times (Section 6.7), and its 95% ranges are slightly too narrow (heavier tails than a Normal distribution). The model gives a sound first estimate, not a precise valuation.

8.2 Data quality issues

- **Asking prices, not sale prices.** Advertisements show what sellers hope to get. Actual transaction prices are usually lower and are not public in Sri Lanka.

- **Errors in the source data:** a price-scale error (Section 3.2), 198 duplicates (Section 3.3) and 21 implausible prices (Section 3.4) had to be corrected or removed.
- **Repeated developer listings.** Many plots from the same block, advertised at the same price, can appear in both the training and test sets. This makes the seen-area scores somewhat optimistic for genuinely new developments.
- **Out of date.** The data is from mid-2021. The 2022 economic crisis and high inflation have changed land prices considerably, so rupee values are out of date, though the relative patterns are more likely to hold.
- **Missing plot features** such as road frontage, road width, shape, slope and flood risk are known to affect Sri Lankan land prices strongly but are not in the dataset.

8.3 Risks of bias or unfairness

- **Uneven coverage.** 88 of the 160 areas have fewer than 10 listings, while a few suburbs (Ranala, Makandana, Kaduwela) have hundreds. Predictions for under-represented areas are less reliable. NGBoost partly makes this visible: its ranges are about 23% wider for areas with fewer than 10 listings (Section 6.7), so users of the application can see the extra uncertainty instead of receiving a falsely precise number.
- **Over-representation of developer blocks.** Suburban plots sold in batches by developers dominate the data, so the model reflects developers' pricing more than private sellers'.
- **Reinforcing existing inequality.** The model learns that central, well-served areas are expensive and outer areas cheap. Used to set prices, it could reinforce these gaps, for example by systematically undervaluing land in less developed areas.
- **Asking-price bias.** If some sellers systematically over-ask (or under-ask), the model inherits that bias.

8.4 Potential real-world impact and ethical considerations

- **Positive impact:** a free, transparent first estimate could help ordinary buyers and sellers judge whether an advertised price is reasonable, in a market with no public price index.
- **Risk of misuse:** the model must not be used for official valuations, loan or mortgage decisions, tax assessment or compensation, because it is trained on 2021 asking prices with area-level location only. The application states this ("a rough guide, not a valuation").
- **Price anchoring:** widely used price estimates can pull the market towards themselves. Showing a range rather than a single number (as the application does with NGBoost) and the data date reduces this risk.
- **Transparency:** the application explains every prediction (Section 7.2), so users can see *why* a price was suggested rather than trusting a black box.
- **Privacy:** the data contains no personal information (Section 2), and the project uses public data only.

8.5 Future work

Per-plot coordinates, recent (post-2022) prices, actual sale prices and plot features such as road access would improve the model more than any further tuning. Regular retraining through the existing pipeline (Section 7.3) would keep it up to date.

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9. Conclusion

A model was built to estimate the asking price per perch of land in the Colombo district from 4,275 cleaned listings. Preprocessing corrected a price-scale error in the data, removed duplicates, and handled the heavily skewed target with a log transform, so that only 21 true outliers had to be removed instead of 297 genuine high-value listings. Random Forest and XGBoost were tuned instead of used with defaults, which improved R^2 from about 0.83 to 0.86. **NGBoost**, an algorithm not covered in the lectures, was added because it predicts a probability distribution rather than a single number. It ranked third of eight models on cross-validated error (test R^2 0.863), and its 80% price ranges contained 81.9% of actual test prices. The stacked ensemble was selected as the final model on cross-validation, reaching an R^2 of 0.866 on the log scale and a median error of about 12%. The application combines both: the ensemble's price with NGBoost's likely range. A Random Forest classifier assigned Low, Medium and High price bands with 90.5% accuracy.

The most important finding came from the data rather than the models: the location features are recorded per area, not per plot. As a result, **location (area) explains most of the price**, all well-tuned tree models perform almost identically, and accuracy is limited by the data rather than by the choice of algorithm. SHAP confirmed this from inside the model: location accounts for 87% of its decisions. SHAP and LIME explanations of the held-out listing agreed on its main drivers (distance from the city, the area itself, and distance to good schools). Better data, meaning per-plot coordinates, recent prices and plot features such as road access, would improve the model more than any further tuning.

Use of AI Tools

AI assistance was used for coding support and for generating the front-end application.

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